

MATH 3210-004 PSet 10

Specification

Instructor: Alp Uzman

Subject to Change; Last Updated: 2026-04-04 10:20:06-06:00

1 Background

This problem starts off with two problems regarding Henstock-Kurzweil integrals and the Fundamental Theorem of Calculus, and then transitions to the study of higher order derivatives and Taylor polynomials.

Saks Lemma The first problem below is often referred to as the Saks (or Saks-Henstock) Lemma¹. Here is some heuristics: Say $f : K \rightarrow \mathbb{R}$ is HK integrable, for K a compact interval. If we were to take a partition $\mathcal{P}$ of K , then by the additivity of HK integrals over intervals (aka by the cocycle property), the integral of f over K is equal to the sum of the integrals of f over all $P \in \mathcal{P}$, more specifically we have

$$\int_{\text{HK}} f(x) dx = \sum_{P \in \mathcal{P}} \int_{\text{HK}} f(x) dx,$$

where $\text{clos}(P) = [\inf(P), \sup(P)]$ is the **closure** of P ; the reason why we take the integrals over closures is because in our definition of a (finite interval) partition we allow $P \in \mathcal{P}$ to be *any interval*. On the

¹See e.g. §1.11 in the Fonda book.

other hand, if $s : \mathcal{P} \rightarrow K$ is a sampling function adapted to $\mathcal{P}$, we have that

$$R(f; \mathcal{P}, s) = \sum_{P \in \mathcal{P}} f \circ s(P) \ell(P).$$

We know that HK integrability guarantees that for $(\mathcal{P}, s)$ fine enough,

$$R(f; \mathcal{P}, s) \approx_{\text{HK}} \int_K f(x) dx,$$

in other words

$$\left| R(f; \mathcal{P}, s) - \int_K f(x) dx \right| \approx 0.$$

Expanding the two expressions as sums over elements of $\mathcal{P}$, and applying the triangular inequality, we obtain

$$\begin{aligned} & \left| R(f; \mathcal{P}, s) - \int_K f(x) dx \right| \\ &= \left| \sum_{P \in \mathcal{P}} f \circ s(P) \ell(P) - \sum_{P \in \mathcal{P}} \int_{\text{HK}} \int_{\text{clos}(P)} f(x) dx \right| \\ &= \left| \sum_{P \in \mathcal{P}} \left[f \circ s(P) \ell(P) - \int_{\text{HK}} \int_{\text{clos}(P)} f(x) dx \right] \right| \\ &\leq \sum_{P \in \mathcal{P}} \left| f \circ s(P) \ell(P) - \int_{\text{HK}} \int_{\text{clos}(P)} f(x) dx \right|. \end{aligned}$$

Just because the LHS is small, we don't a priori have that the RHS is small. Saks Lemma precisely is about the establishment that indeed HK integrability ultimately implies that *the RHS is small too*, for $(\mathcal{P}, s)$ fine enough. Thus the Riemann sum not only approximates the integral at large, but also each piece of the integral separately.

As a final note, while the Saks Lemma often shows up in the context of HK integrals, it is also valid for Riemann integrals, and Saks initially realized this lemma for **Burkill integrals**, where one associates to each element of a partition an "area", thus generalizing the Riemann sums over summands of the form "height" $\times$ "width".

Taylor Approximation Here is a quick summary of definitions and notations for Taylor approximation; there really is no common consensus regarding notation in the literature. Let I be a nonempty open interval, $f \in F(I; \mathbb{R})$, $x_* \in I$, $n \in \mathbb{Z}_{\geq 0}$. Suppose f is n -times differentiable (with the n -th derivative not necessarily continuous). Then

- The **order n Taylor polynomial of f at x_*** is the following polynomial function:

$$\begin{aligned} T_n(f; x_*) : \mathbb{R} &\rightarrow \mathbb{R}, \\ x &\mapsto \sum_{k=0}^n \frac{f^{(k)}(x_*)}{k!} (x - x_*)^k \\ &= f(x_*) \\ &\quad + f'(x_*)(x - x_*) \\ &\quad + \frac{f''(x_*)}{2} (x - x_*)^2 \\ &\quad + \cdots \\ &\quad + \frac{f^{(n)}(x_*)}{n!} (x - x_*)^n. \end{aligned}$$

Note that $T_n(f; x_*)$ is a polynomial function of degree *at most* n , and in general it *depends* on the base point x_* , just like the equation of the line tangent to the graph of f at x_* , which is precisely $T_1(f; x_*)$.

- The **order n Taylor remainder of f at x_*** is the following function:

$$\begin{aligned} R_n(f; x_*) : \mathbb{R} &\rightarrow \mathbb{R}, \\ x &\mapsto f(x) - T_n(f; x_*)(x). \end{aligned}$$

Since $R_n(f; x_*)$ is the difference of two n -times differentiable functions, it too is differentiable.

2 Generative AI and Computer Algebra Systems Regulations

This section applies only if you choose to use either a **generative AI tool** (e.g. chatbots) or a **computer algebra system (CAS)** while working on this problem set. If you do not use such tools, you may skip this section.

The use of these tools while completing this problem set is permitted, provided it is done responsibly and in a manner that supports your learning. Note however that in exams you will not be allowed to use any AI or CAS.

2.1 Disclosure

If you use such tools, you must disclose this fact in the designated section of the form you will complete as part of your submission. You must also share your reflections on which parts of the problem required your own mathematical judgment, insight, or decision-making, and which parts could reasonably be delegated to a computational or generative tool.

2.2 Guidelines for Responsible Use

If you use a generative AI tool or chatbot, you must adhere to the following guidelines:

- During the chat you may share with the chatbot parts or all of this specification document, as well as parts of the textbook or other sources.
- Directly asking the tool for complete problem solutions is prohibited.
- Appropriate uses include asking for conceptual explanations, checking intermediate steps, clarifying definitions or theorems, or exploring alternative solution approaches, provided that the final work submitted is your own.
- Regardless of the tools used, you are expected to understand, justify, and be able to reproduce all submitted work independently. The use of AI or CAS does not reduce or replace this expectation.

3 What to Submit

Submit your detailed solutions to each of the problems below. While the prompts may seem lengthy, the additional text is intended to guide you by providing context and helpful details.

When tackling statements in MATH 3210, follow these four steps (in no particular order):

1. **Assess the truth of the statement:** Determine, either exactly or probabilistically, whether the statement is true.

2. **Build a case:** Come up with examples or begin drafting a proof to identify potential issues or areas where the argument may break down.
3. **Construct a proof:** Provide a formal argument to support your conclusion.
4. **Perturb the statement:** Experiment with logical variations of the statement. Can you prove a stronger/weaker result, or find alternative proofs?

In this problem set, and in later problem sets as well as exams, your work must address the third step (aka constructing a proof). While work regarding the remaining three items likely will help you write a formal proof, understand the material as well as identify connections between different concepts, you do not need to turn in generalizations or multiple proofs of the same statement, nor do you need to provide examples or counterexamples (unless you are specifically asked to). However, demonstrating these steps can contribute to partial credit when appropriate.

Statements in problems will typically be presented in neutral language. For example, a problem may simply state "P" (for P a well-defined statement) rather than "Show that P" or "Prove that P is false". Your task is to determine the truth of the statement and provide a formal proof or refutation.

If a statement is false, it is highly recommended to propose a corrected version of the statement and prove this corrected version. This process, part of "perturbing the statement", will enhance your mathematical **error detection and correction skills**.

Unless explicitly stated otherwise, all problems require proofs. If you provide an example, you must also prove that the example satisfies the necessary conditions.

Some problems may be (variants of) theorems or exercises from the textbook. Such correspondences will not be mentioned; it'll be up to you to locate them (if need be)!

Make sure that each solution is properly enumerated and organized. Start the solution to each problem on a new page, and consider using headings or subheadings to structure your work clearly. This will not only aid in your thought process but also ensure that no part of your solution is overlooked during grading.

1. This problem is a prequel to the next problem. Let I be a nonempty open interval, $f \in F(I; \mathbb{R})$, $K \subseteq I$ be a compact subinterval. Then

(a) The following are equivalent:

I: $f \in HK(K; \mathbb{R})$.

II: For any subinterval $L \subseteq K$, there is a real number $\mathfrak{I}(L)$ with the property that for any $\varepsilon \in \mathbb{R}_{>0}$, there is a $\delta \in \mathcal{G}(K)$ such that for any sampled partition $(\mathcal{P}, s)$ of K , if $(\mathcal{P}, s)$ is δ -fine, then

$$\sum_{P \in \mathcal{P}} |f \circ s(P) \ell(P) - \mathfrak{I}(P)| < \varepsilon.$$

(In this case $\mathfrak{I}(P) = {}_{HK} \int_{\text{clos}(P)} f(x) dx$.)

III: For any subinterval $L \subseteq K$, there is a number $\mathfrak{I}(L) \in \mathbb{R}$ with the property that for any $\varepsilon \in \mathbb{R}_{>0}$, there is a $\delta \in \mathcal{G}(K)$ such that for any sampled partition $(\mathcal{P}, s)$ of K , if $(\mathcal{P}, s)$ is δ -fine, then for any $\mathcal{P}' \subseteq \mathcal{P}$:

$$\sum_{P \in \mathcal{P}'} |f \circ s(P) \ell(P) - \mathfrak{I}(P)| < \varepsilon.$$

(b) If $f \in HK_{\text{loc}}(K; \mathbb{R})$, then for any $\varepsilon \in \mathbb{R}_{>0}$, there is a $\delta \in \mathcal{G}(K)$

such that for any sampled partition $(\mathcal{P}, s)$ of K , if $(\mathcal{P}, s)$ is δ -fine, then

$$\left| R(|f|; \mathcal{P}, s) - \sum_{P \in \mathcal{P}} |\mathfrak{J}(P)| \right| < \varepsilon,$$

where

$$\mathfrak{J}(P) = \int_{HK} \int_{\text{clos}(P)} f(x) dx.$$

2. This problem is a sequel to the previous problem. Let I be a nonempty open interval, $f \in HK_{\text{loc}}(I; \mathbb{R})$. Then

- (a) For any $\gamma \in I$, $F_{HK, \gamma} \in F(I; \mathbb{R})$ defined by $x \mapsto \int_{HK, \gamma}^x f(u) du$ is continuous.
- (b) $F_{HK, \bullet} \in F(I; C^0(I; \mathbb{R}))$ defined by $\gamma \mapsto F_{HK, \gamma}$ is continuous relative to the C^0 metric.

3. There is nothing to prove in this problem. Consider the following functions $f : \mathbb{R} \rightarrow \mathbb{R}$:

I: $x \mapsto x/(x^2 + 1)$

II: $x \mapsto \cos(1 + x^2)$

III: $x \mapsto (x^2 + 1)^{25}$

For each of the given functions f ; compute $a = (a_0, a_1, a_2, a_3) \in \mathbb{R}^4$ such that for any $x \in \mathbb{R}$,

$$T_3(f; 1)(x) = a_0 + a_1x + a_2x^2 + a_3x^3.$$

4. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $x \mapsto \begin{cases} \exp(-1/x^2) & , \text{ if } x \neq 0 \\ 0 & , \text{ if } x = 0 \end{cases}$. Then

- (a) For any $n \in \mathbb{Z}_{\geq 1}$, $f(x) = o_{x \rightarrow 0}(x^n)$.
- (b) f is differentiable at $x = 0$ and $f'(0) = 0$.
- (c) For any $n \in \mathbb{Z}_{\geq 0}$, there is a polynomial function P_n of degree $3n$ such that the n -th derivative of f satisfies the formula

$$\forall x \in \mathbb{R} \setminus \{0\} : f^{(n)}(x) = P_n\left(\frac{1}{x}\right) f(x).$$

- (d) For any $n \in \mathbb{Z}_{\geq 0}$, $f^{(n)}(0)$ exists and is zero.
- (e) For any $n \in \mathbb{Z}_{\geq 0}$, $T_n(f; 0)$ is the function that is constantly zero.
- (f) $f \in C^\infty(\mathbb{R}; \mathbb{R})$ (aka, f is **infinitely differentiable**).

5. Let $p \in \mathbb{Z}_{\geq 0}$ and $f \in F(\mathbb{R}; \mathbb{R})$ be p -times differentiable. Then

- (a) For any $x_* \in \mathbb{R}$ and for any $q \in \{0, 1, \dots, p\}$:

$$T_p(f; x_*)^{(q)} = T_{p-q}(f^{(q)}; x_*).$$

- (b) f is a polynomial function of degree at most p iff

$$\forall x_1, x_2 \in \mathbb{R} : T_p(f; x_1) = T_p(f; x_2).$$

4 How to Submit

- **Step 1 of 2:** Submit the form at the following URL:

<https://forms.gle/MzUfKRFzWQ9b1qjT7>.

You will receive a zero for this assignment if you skip this step, even if you submit your work on Gradescope on time.

- **Step 2 of 2:** Submit your work on Gradescope at the following URL:

[https://www.gradescope.com/courses/1212058/assignments/7390772,](https://www.gradescope.com/courses/1212058/assignments/7390772)

see the Gradescope [documentation](#) for instructions.

5 When to Submit

This problem set is due on April 10, 2026 at 11:59 PM.

As per the course's syllabus, late submissions, up to 24 hours after this deadline, will be accepted with a 10% penalty. Submissions more than 24 hours late will not be accepted unless you contact the course staff with a valid excuse before the 24-hour extension expires.